Introduction to STEM Pathways and Careers

Overview

Introduction to STEM Pathways and Careers was developed for the Louisiana Department of Education through a partnership between Louisiana State University and the East Baton Rouge Parish School System, with partial funding from the National Science Foundation. The high school credit course serves as a basic career readiness course on all Jumpstart 2.0 Pathways including the STEM Pathways, and has successfully been offered at 7th and 8th grade in the past. This year long project-based course explores four main pathways of STEM education and possible careers in the fields of 1) Computational Thinking and Computer Science, 2) Pre-Engineering 3) Digital Design and Emergent Media, and 4) Biomedical Sciences, now Pre-Veterinary and Pre-Health. The course also prepares students for the CIW Internet Business Associate certification exam. This course is meant to expose students to these overarching concepts:

- To expand awareness of various careers and occupational pathways related to STEM
- To stimulate the understanding of higher order thinking processes such as the engineering design process, the scientific method, and computational thinking.
- To develop foundational knowledge and skills in the JumpStart pathways and careers as related to STEM and utilize the knowledge and skills in their current educational setting.
- To increase interest in the four core areas of STEM related to this class through project-based activities that are also standards based.
- To prepare students for the CIW Internet Business Associate certification exam.

Objectives

- Identify career opportunities in various STEM fields.
- Use computing technology for creative expression.
- Explain the Software Development Process through projects and activities where students design and code.
- Explain the Engineering Design Process and utilize the process in various engineering projects.
- Describe the role of visual communication and its cultural implications in society.
- Use various forms of digital and emergent media to design their own products.
- Explore and analyze medical scenarios in the areas of forensics, comparative anatomy, ecology studies, biochemistry, microbiology, and bioinformatics.

Assessment Students will be assessed using projects throughout the class, exams, and daily/ weekly assignments.

Course Essentials

Equipment	Cost/Unit
Software	$0 (Required) or up to $500 for paid premium software that more options.
Computer or laptop	
Other Materials	Reusable: $1500 Consumable: (up to $500 per year, replace as needed.)

First Semester Course Outline

Unit 1: Introduction to Pre-Engineering	Engineering design process, Careers, Ethics, Professionalism, Circuits, Robotics, Build an Arduino Robot, Build a Better Mousetrap Car, Web browsing and hardware/ software, Business email, Email systems, and online communication
Unit 2: Digital Design and Emergent Media	What is DDEM, Careers, Design Principles, Website Creation, Digital Portfolios, Web Design, Digital Animation, 3D Design, Graphic Design for a Business, Digital Storytelling. Digital Communication and Technology, Active Web content

Second Semester Course Outline

Unit 3: Biomedical Sciences	What are Biomedical Sciences, Careers, The Scientific Method, Forensics and Comparative Anatomy, Ecology and Coastal Studies, Biochemistry and Microbiology, Bioinformatics. Digital Communication: Internet Technology, Databases, Search Engines, and Research, Midterm
Unit 4: Introduction to Computer Science and Computational Thinking Overview	Why Computing, Algorithms and Sequencing, Debugging, Intro to Scratch, Event Programming, Sprite Conversations, Broadcasting Events, Conditional Events, Data Storage, Variables and Conditionals, Making a Quiz, Reverse Engineering, Hackathon, Intro to HTML Online Safety, Internet Resource Tools, IT Projects. Career Capstone

INTRODUCTION TO STEM PATHWAYS AND CAREERS

1. Materials
   Internet access, one-to-one computer use daily (all materials are accessible via Chromebooks), and access to the LSU servers.

Reusable Hardware/Material	Recommended Unit	Cost/Unit
Various reusable material and hardware for projects	1 per classroom	$1,500
Consumables		
Various consumables for projects	1 per classroom	$500

*Complete supply list can be found here.

2. Required software, networking access, and access to LSU servers
   - Students will need to sign up with individual accounts for online development environments: scratch.mit.edu, tinkercad.com, canva.com, and code.org.
   - Students will need access to YouTube instructional videos relevant to the course, as well as other educational video repositories.
   - Teachers will need to be able to access the LSU servers using several Internet protocols including but not limited to HTTPS and SSH.
   - Students and teachers will access the curriculum and teaching materials through Google Drive.
   - Principals will need to communicate with the district's information technology department to ensure that there are no technological restrictions that block access to the LSU servers in the lsu.edu, college-readiness.lsu.edu, louisianatrainingplatform and pathways.lsu.edu domains on any port. In addition to the sites mentioned above, students will need web access to various other sites. A complete whitelist can be found here.
   - **PLEASE NOTE:** If access to certain sites and programs is prohibited, suggestions for alternative lessons and activities are provided within the curriculum.

3. Requirements for IBC Alignment: **CIW Internet Business Associate**
   For schools who are pursuing the aligned IBC for this course
   a. Schools must have the appropriate testing software installed on the testing computers. For this credential the testing software is through CIW. More information can be found here: https://ciwcertified.com/exam-requirements/
   b. Some industry based credential providers also provide practice tests softwares such a GMetrix which may be installed. Please check with individual schools if they plan to use a practice software.
   c. Schools must also typically register as an approved testing center prior to being allowed to proctor the exams.
   d. Teachers must have earned the IBC prior to students sitting for the exam. Once they have passed the credential, they need to be set up as a proctor in the CIW system which must be done by the CIW team. Please contact your regional CIW representative.
   e. Teachers should report to the LSU Point of Contact their overall pass rates (i.e. 15/20 students passed)

4. Required teacher collaborations
   Teachers will communicate with LSU instructors via emails and apps hosted on the LSU servers.
5. Required administration of course content, pre/post test, and research instruments
   All required materials and instruments will be either posted in the LSU servers, Google Drive, or their location announced via email or on Google drive.
7. Course Work
   Teachers must present the course material in sequence or as approved by collaboration with the LSU Pathway Point-of-Contact. Teachers are expected to deliver a minimum of 80% of the course material for course fidelity and for the opportunity for IBC attainment (where applicable).
6. Other
   As this is a project-based learning class, we strongly suggest that each section of the course be limited to a *maximum* of 25 students. The course is dependent on the teacher providing feedback and reviewing student code. The course requires that teachers have adequate time to interact with each student.